

Task: CRUD Operations using SQLite3 (Python)

Practice all four database operations — Create, Read, Update, Delete — using Python's built-in `sqlite3` module.

Objective

Build a small Python program that connects to a SQLite database named **students.db** and performs all CRUD operations on a **students** table.

Task 1 — Setup: Create Database & Table

- Connect to (or create) a database file called **students.db**.
- Create a table named **students** with the following columns:
 - id (INTEGER, PRIMARY KEY, AUTOINCREMENT)
 - name (TEXT, NOT NULL)
 - age (INTEGER)
 - course (TEXT)

Task 2 — CREATE: Insert Records

- Insert a single student record using **cursor.execute()** with placeholders (?).
- Insert at least 3 more student records in one go using **cursor.executemany()** with a list of tuples.
- Commit the changes after inserting.

Task 3 — READ: Query Records

- Fetch and print all rows from the **students** table using **fetchall()**.
- Fetch and print only one row using **fetchone()**.
- Write a query to fetch only students whose **age > 21**.
- Write a query to fetch only students enrolled in a specific **course** (e.g. "BCA").

Task 4 — UPDATE: Modify Records

- Update the **course** of a student with a specific **id** (use a placeholder, not a hardcoded value).
- Update the **age** of a student by **name**.
- After each update, commit and re-query the table to confirm the change.

Task 5 — DELETE: Remove Records

- Delete a student record by **id**.
- Delete all students belonging to a specific **course**.
- Confirm deletion by printing the table before and after.

Bonus (Optional)

- Wrap each CRUD operation in its own function: `create_student()`, `read_students()`, `update_student()`, `delete_student()`.
- Add a try/except block around your database operations to handle errors (e.g. `sqlite3.IntegrityError`).
- Use **`conn.close()`** at the end of the script to close the connection properly.
- Print a simple text-based menu (Insert / View / Update / Delete / Exit) and let the user choose an operation in a loop.

Reminder: Always use `?` placeholders for values in SQL queries instead of f-strings or string concatenation, and call **`conn.commit()`** after every INSERT, UPDATE, or DELETE.

Deliverable Checklist

#	Operation	Done?
1	Create database + students table	■
2	Insert one record (execute)	■
3	Insert multiple records (executemany)	■
4	Read all records (fetchall)	■
5	Read one record (fetchone)	■
6	Read with WHERE condition(s)	■
7	Update record by id	■
8	Update record by name	■
9	Delete record by id	■
10	Delete records by condition	■